

Non-Sigma-Unital Coronas Are Universal for Unital C^* -Algebras

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

The previous packet

2311.14257_sigma_unital_coronas_fail_bcp

proved that if A is nonunital and σ -unital, then $\mathcal{M}(A)/A$ fails BCP. The following theorem shows that the σ -unital hypothesis is essential.

Theorem 1. *Let B be a nonzero unital C^* -algebra. There is a nonunital non- σ -unital C^* -algebra A such that*

$$\mathcal{M}(A)/A \cong B.$$

Indeed, one can take

$$A = C_0([0, \omega_1), B).$$

Corollary 1. *There is a nonunital non- σ -unital C^* -algebra A such that $\mathcal{M}(A)/A$ has UBCP. For example,*

$$A = C_0([0, \omega_1))$$

satisfies

$$\mathcal{M}(A)/A \cong \mathbb{C}.$$

Idea of the Proof

The ordinal space $[0, \omega_1)$, with the order topology, is locally compact and non- σ -compact. Its decisive feature here is that every continuous map from $[0, \omega_1)$ to a metric space is eventually constant. Since a C^* -algebra is a metric space in norm, every bounded continuous B -valued function on $[0, \omega_1)$ has an eventual value in B . For unital B ,

$$\mathcal{M}(C_0([0, \omega_1), B)) = C_b([0, \omega_1), B),$$

and the ideal $C_0([0, \omega_1), B)$ is precisely the kernel of the eventual value map. The corona is therefore B .

Proof

Let

$$X = [0, \omega_1)$$

with its order topology.

Lemma 1. *If Y is a metric space and $f : X \rightarrow Y$ is continuous, then f is eventually constant.*

Proof. Fix $\varepsilon > 0$. We first show that there is $\alpha_\varepsilon < \omega_1$ such that

$$\text{diam } f([\alpha_\varepsilon, \omega_1)) < \varepsilon.$$

If not, start with $\alpha_0 = 0$. Having chosen α_n , choose ordinals $\beta_n, \gamma_n \geq \alpha_n$ such that

$$d(f(\beta_n), f(\gamma_n)) \geq \varepsilon.$$

Then choose α_{n+1} strictly larger than both β_n and γ_n . Let

$$\delta = \sup_n \alpha_n.$$

This supremum is still a countable ordinal, hence $\delta < \omega_1$. Since both β_n and γ_n eventually lie above every α_N , both sequences converge to δ in the order topology. By continuity, $f(\beta_n) \rightarrow f(\delta)$ and $f(\gamma_n) \rightarrow f(\delta)$, contradicting

$$d(f(\beta_n), f(\gamma_n)) \geq \varepsilon$$

for all n .

Thus for every $m \geq 1$ there is $\alpha_m < \omega_1$ such that

$$\text{diam } f([\alpha_m, \omega_1)) < 1/m.$$

Let $\alpha = \sup_m \alpha_m$. Again $\alpha < \omega_1$. If $\xi, \eta \geq \alpha$, then $d(f(\xi), f(\eta)) \leq 1/m$ for every m , so $f(\xi) = f(\eta)$. Hence f is constant on $[\alpha, \omega_1)$. $\square$

Lemma 2. *Let B be a unital C^* -algebra. Then*

$$\mathcal{M}(C_0(X, B))/C_0(X, B) \cong B.$$

Proof. Since B is unital, the standard multiplier identification gives

$$\mathcal{M}(C_0(X, B)) = C_b(X, B),$$

with pointwise operations and norm. By Lemma 1, each $f \in C_b(X, B)$ is eventually constant. Define

$$\rho : C_b(X, B) \longrightarrow B$$

by letting $\rho(f)$ be this eventual value. The map ρ is a surjective $*$ -homomorphism, since every $b \in B$ is the eventual value of the constant function $x \mapsto b$.

Its kernel is exactly $C_0(X, B)$. Indeed, if $f \in C_b(X, B)$ has eventual value 0, then f is eventually zero, so its support is contained in $[0, \alpha]$ for some $\alpha < \omega_1$; this interval is compact, hence $f \in C_0(X, B)$. Conversely, if $f \in C_0(X, B)$ had nonzero eventual value $b \neq 0$, then for $0 < \varepsilon < \|b\|$ the set

$$\{x \in X : \|f(x)\| \geq \varepsilon\}$$

would contain a tail $[\alpha, \omega_1)$, which is not compact. This contradicts $f \in C_0(X, B)$. Therefore

$$C_b(X, B)/C_0(X, B) \cong B.$$

□

Lemma 3. *If $B \neq 0$ is unital, then $A = C_0(X, B)$ is nonunital and not σ -unital.*

Proof. The space $X = [0, \omega_1)$ is locally compact and noncompact, so $C_0(X, B)$ is nonunital.

We next recall that X is not σ -compact. Every compact subset of $[0, \omega_1)$ is bounded by some countable ordinal. A countable union of such bounded compact sets is still bounded below ω_1 , and therefore cannot cover X .

Suppose, toward a contradiction, that $C_0(X, B)$ were σ -unital. Then it has a countable positive contractive approximate unit (u_n) . The compact sets

$$K_n = \{x \in X : \|u_n(x)\| \geq 1/2\}$$

cover X . Indeed, fix $x \in X$. Choose $g \in C_c(X)$ with $g(x) = 1$ and choose $b \in B$ with $\|b\| = 1$. Put

$$a(t) = g(t)b \in C_0(X, B).$$

For some n , $\|u_n a - a\| < 1/2$. Evaluating at x gives

$$\|u_n(x)b - b\| < 1/2,$$

and hence $\|u_n(x)\| \geq 1/2$.

Since each function $x \mapsto \|u_n(x)\|$ belongs to $C_0(X)$, each K_n is compact. Thus $X = \bigcup_n K_n$ would be σ -compact, a contradiction. Therefore A is not σ -unital. □

Proof of Theorem 1. Take $A = C_0(X, B)$. Lemma 3 says that A is nonunital and not σ -unital. Lemma 2 gives

$$\mathcal{M}(A)/A \cong B.$$

□

Proof of Corollary 1. Apply Theorem 1 with $B = \mathbb{C}$. The one-dimensional space $\mathbb{C}$ has UBCP, so the corona $\mathcal{M}(C_0([0, \omega_1)))/C_0([0, \omega_1))$ has UBCP. □

Consequences for Question 3

The sigma-unital corona theorem is sharp in a strong sense. Sigma-unital nonunital coronas all fail BCP by the annulus construction, but non- σ -unital coronas do not have a uniform BCP behavior: they can be made isomorphic to any prescribed unital C^* -algebra. Therefore the remaining arbitrary C^* -algebra classification cannot be reduced to a blanket statement that coronas fail BCP.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with non- σ -unital corona, multiplier corona, $[0, \omega_1)$, ordinal space, eventual constancy, BCP, and UBCP. Existing packets contained the sigma-unital corona failure theorem and the earlier unital stabilized-corona obstruction, but no packet recorded this non- σ -unital universality phenomenon.

The proof uses only standard facts about $C_0(X, B)$, multiplier algebras for unital coefficient algebras, and the elementary topology of the ordinal space $[0, \omega_1)$.

Human Review Focus

Check Lemma 1, especially the diagonal sequence contradiction for tail diameters, and Lemma 2, where $C_0(X, B)$ is identified as the kernel of the eventual-value map. Those two lemmas carry the theorem.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.